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Optimal return shipping insurance policy with consumers' anticipated regret

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Abstract

We study return shipping insurance (RSI) policies prevalent on platforms such as JD.com and Taobao.com. Retailers on those platforms can purchase and provide RSI for consumers (RRSI) or offer an option for consumers to buy RSI themselves (CRSI). With either RSI, consumers will be partially compensated by an insurer for their shipping fees associated with product returns. As the consumers' uncertainty about product fit is realized only after purchase, their decisions whether to purchase CRSI may lead to postpurchase regret. Considering these anticipated regret behaviors, we investigate the optimal RSI policy for a monopolistic online retailer and an insurer. We show that the retailer offers RRSI only if the retailer's return handling cost is relatively low and consumers' return shipping cost is in an intermediate range; otherwise, the consumers' strong propensity for uninsured regret may stimulate them to purchase CRSI. Under the optimal RRSI policy, the retailer always charges a higher product price than under no RSI, and surprisingly the consumer demand could expand. In contrast, under the optimal CRSI policy, the retailer always sets a lower product price, but resulting in consumer demand shrinkage. Counterintuitively, CRSI may become a "win-win-win" policy for the retailer, insurer, and consumers.

KEYWORDS
anticipated regret, behavioral economics, pricing, product returns, return shipping insurance

1 | INTRODUCTION

Online retailing has witnessed rapid growth in recent years (eMarketer, 2017). However, many consumers still hesitate to purchase experience goods such as clothes and shoes online because they cannot try the products before purchasing to alleviate their uncertainty about product fit (eMarketer, 2016; Hong & Pavlou, 2014). As a result, product returns have become the new normal. According to Invesp's statistics on online return rates, at least 30% of all products ordered online are returned, much higher than 8.89% in brick-and-mortar stores (Spillane, 2022). U.S. return delivery costs have been steadily increasing from $350 billion to $550 billion from 2017 to 2020 (Placek, 2022).

Most online retailers offer lenient return policies to drive sales (Abdulla et al., 2019). In China, online retailers on platforms such as Taobao.com and JD.com have long been offering 7-day "no-question asked" full-refund return policies (Millward, 2014). In addition, the European Union obliges online firms to offer a no-questions return period of 14 days (Economist, 2013). Nevertheless, 100% money-back guarantees (MBGs) do not mean online product returns are hassle-free for consumers (Davis et al., 1998). In general, consumers are responsible for the return shipping fee, unless the products are defective (Bower & Maxham, 2012). As the return shipping fee is not cheap, it impedes online purchasing due to the high uncertainty of mismatch (Lawton, 2008). A survey conducted by Invesp shows that 79% of consumers want free return shipping, but only around 49% of retailers currently offer that policy (Spillane, 2022). Recently, we also

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TABLE 1 Summary of the survey.

Statement	Respondents agreed
(A.1) The return shipping fee does not play a role in my online purchase decisions.	6.35%
(A.2) The RSI (RRSI or CRSI) does not play a role in my online purchase decisions.	6.56%
(B.1) If I purchase CRSI but then receive a fit product that does not need to return, I do not feel regret that I have purchased CRSI (i.e., no insured regret).	20.00%
(B.2) If I do not purchase CRSI but then receive a misfit product that has to return, I do not feel regret that I have not purchased CRSI (i.e., no uninsured regret).	3.35%
(C.1) The uninsured regret is stronger than the insured regret. (Note: only for respondents who have insured/uninsured regret.)	44.44%
(C.1.1) I feel the uninsured regret is stronger than the insured regret because the CRSI premium is cheap compared with the high compensation. (Note: only for respondents who feel the uninsured regret is stronger than the insured regret)	67.21%

Abbreviations: RSI, return shipping insurance; RRSI, retailer-RSI; CRSI, customer-RSI

conducted a survey[1] using the popular tool SoJump and collected 976 responses. The main results are summarized in Table 1. The survey shows that only for 6.35% of our respondents, the return shipping fee does not play a role in their product purchase decisions.

In response to consumer desire for free return shipping, major Chinese online platforms (e.g., Taobao.com, Tmall.com, and JD.com) adopt return shipping insurance (RSI), an innovative insurance tool, to alleviate consumers' concerns about product returns (Sheng et al., 2016). The insurer will compensate consumers with RSI for a prespecified amount, up to the actual shipping fee, in the case of product returns. Initially, Taobao.com launched RSI through Huatai Insurance Group[2] to alleviate the retailers' and consumers' risks and costs associated with online returns (He, 2013). Witnessing the success of RSI, other online platforms in China, including JD.com and Suning.com, have eagerly embraced RSI policies. Now, more than 11 insurers sell RSI services, including China Life,[3] China Pacific Insurance,[4] and ZhongAn Insurance.[5] A report from the Insurance Association of China shows that the revenue from RSI has been growing rapidly since 2016, which has become the highest among non–auto insurance category (Tu & Zhang, 2020). In 2020, Taobao.com announced that consumers from 17 overseas countries and regions (Hong Kong, Macao, and Taiwan), including the United States, can enjoy RSI when shopping on the platform (Tencent, 2020). Indeed, RSI is a key factor in consumers' purchasing decisions (for 93.44% of respondents in our survey).

There are two types of RSI policy: (1) the retailer-RSI (RRSI) policy under which online retailers purchase RSI from insurance companies and provide it for all consumers for free and (2) the customer-RSI (CRSI) policy under which the retailer offers an option for consumers to purchase RSI directly from the insurer during the merchandise checkout stage. For RRSI, the retailer buys RSI for the consumers, reducing their potential product return shipping cost. In contrast, under the CRSI policy, consumers must pay for RSI themselves, and thus retailers actually transfer the premium cost to consumers. Irrespective of the RSI type, it is harder for insurers to gain profitability if more consumers tend to return products, resulting in more compensation from the insurer.

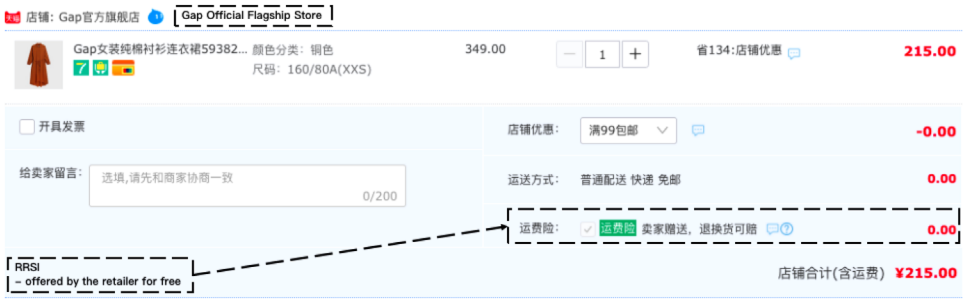
One may anticipate that retailers would prefer the CRSI policy to the RRSI policy, as consumers pay the CRSI premium themselves. However, most retailers adopt the RRSI policy (Li et al., 2021, 2023; Xiong & Liu, 2023; Zhang et al., 2022). Why do these retailers not transfer the insurance premium cost to consumers by using CRSI? More interestingly, retailers may adopt different RSI policies even when they sell the same product. Figure 1 shows that both the official flagship store of GAP on Tmall.com[6] and a small seller on Taobao.com[7] sell the same dress of the GAP brand, but the former offers RRSI and the latter offers CRSI. What motivates those retailers to adopt different RSI policies?

Consumers are uncertain about their true needs at the time of purchase; therefore, they may have ex post regret about their purchase decisions of RSI under the CRSI policy (Bell, 1982; Braun & Muermann, 2004). For example, if an individual does not buy RSI but receives a mismatched product and wants to return it, the individual would experience some additional disutility of not having purchased RSI ex ante. Similarly, if an individual does buy RSI but receives a fit item, the individual would regret having purchased RSI ex ante. Our survey shows that more than 80% of respondents have experienced regret about their CRSI purchasing decisions, which reveals that regret aversion plays an important role in consumers' purchasing decisions under RSI policy. Consumer regret behaviors have attracted much attention in academia and the magnitude of consumer regret can be pronounced (Simonson, 1992; Zeelenberg, 1999), especially in the insurance industry (Braun & Muermann, 2004; Fujii et al., 2016; Huang et al., 2008; Wong, 2012). Anticipating possible postpurchase regret, consumers will take it into consideration during the purchasing process (Zeelenberg et al., 1996).

In the presence of consumer regret behaviors in online shopping, the following research questions arise naturally: (1) When is it profitable for an insurer and a retailer to adopt the RSI policy? RRSI or CRSI? (2) How do consumers' regret behaviors shape the RSI policy? (3) Will the RSI policy always increase the consumer surplus?

To address these research questions, we develop an analytical framework. We study a market with a monopolistic online retailer and focus on how the consumers' anticipated regret behavior affects the RSI policy. Consumers are heterogeneous in their knowledge about whether the retailer's product will fit them. The *informed* consumers know their needs exactly before purchasing the product. In contrast, the *uninformed* consumers are uncertain about the product's fit at the time of purchase. The retailer chooses whether to

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(a) RRSI on Tmall.com



(b) CRSI on Taobao.com

FIGURE 1 Return shipping insurance (RSI) examples of GAP. [Color figure can be viewed at wileyonlinelibrary.com]

purchase RSI from an insurer and offer an RRSI policy such that the consumers' potential return shipping fee will be partially paid by the insurer. Otherwise, the retailer considers adopting the CRSI policy, under which uninformed consumers can purchase RSI from the insurer themselves. Meanwhile, the uninformed consumers anticipate that they may ex post experience regret about their RSI purchase decision under the CRSI policy. In line with the behavioral literature, we consider two types of consumer regret: aversion to the regret from not buying RSI, that is, *uninsured regret*, and aversion to the regret from buying RSI, that is, *insured regret*. For either RSI policy, the insurer determines the insurance premium to maximize its benefit.

Our analysis reveals that consumer-anticipated regret behaviors and the heterogeneity in consumer capability of processing information shape the retailer's optimal RSI policy. Either RRSI or CRSI may be optimal, depending on the consumers' regret level. This finding differs from Zhang et al. (2022)'s argument that the RRSI policy is viable if consumers are uncertain about the product quality, but the CRSI policy never benefits the retailer. Note that, our work applies to retailers selling experience goods, such as shoes and clothes, without consumers' concerns about the product quality, while Zhang et al. (2022) focus on the market where "quality is the retailer's private information" and consumers are uncertain about the product quality. Our work explores the viability of the RSI policy through a new lens and provides an alternative explanation for the prevalent RSI policy. Although both types of RSI policy may benefit the retailer as well as the insurer, they impact the retail operations in different ways. RRSI raises the product price, exacerbating the adverse selection problem, but may expand the consumer demand, while the CRSI policy always lowers the product price, mitigates the adverse selection problem, and shrinks the consumer demand.

As a result, the RRSI policy is optimal if the retailer's return handling cost is relatively low and the consumers' return shipping cost is medium. In contrast, a sufficiently strong uninsured regret level calls for the CRSI policy when there is a relatively high return handling cost and intermediate return shipping cost. Although extensive literature (Davis et al., 1995; Huang et al., 2018) has shown the optimality of MBGs, the role of the return shipping cost is underexplored. Taking into consideration the consumers' return shipping fees, we discover that RSI, an insurance tool for possibly reducing consumers' return associated cost, further improves the profit of a retailer that offers MBG. In other words, our findings suggest that a new return policy, that is, "MBG + RSI," is even better than the commonly adopted MBG return policy. Moreover, we show that RSI also benefits retailers who adopt partial-refund policies, such as charging a restocking fee. Interestingly, we find that CRSI may result in a *"win-win-win"* outcome for the retailer, insurer, and consumers.

Our findings potentially explain the retailers' RSI policies shown in Figure 1. Major retailers in online platforms can handle returned products at a low cost because returned products can be cheaply repackaged due to economies of scale and resold in widespread outlets. In contrast, it costs more for small sellers to handle product returns. Therefore, major retailers favor RRSI, while small retailers adopt CRSI.

The remainder of this paper is organized as follows. In Section 2, we discuss the relevant literature and highlight our contribution. The modeling framework is laid out in

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Section 3. We present the results for cases without and with anticipated regret in Sections 4 and 5, respectively. We consider several model extensions in Section 6 and Section EC.1 in the Supporting Information. Finally, Section 7 concludes our work and provides directions for future research.

2 | RELATED LITERATURE

Our study lies at the confluence of four streams of prior research: product return policy, RSI, add-on pricing, and anticipated regret. In this section, we provide an overview of these related works and highlight our contribution.

Our paper builds on prior theoretical literature on product returns. A main research question in this stream is about the optimal refund amount for product returns: full, partial, or none. The full-refund policy is referred to as MBG and is assumed to signal the product quality. For example, Che (1996), McWilliams (2012), and Moorthy and Srinivasan (1995) compare MBG with the no-refund policy in a monopoly or duopoly setting. More recently, Huang et al. (2018) discover that both retailers in a competitive market should adopt MBGs, but MBGs do not necessarily favor the low-quality retailer. Huang and Huang (2021) examine MBGs in a distribution channel consisting of one manufacturer and two retailers with different qualities. Huang et al. (2021) study the pricing and product return policies when physical and online stores compete for consumers who can visit the physical store before buying online. They find that the online store should offer MBGs when the salvage advantage outweighs the total return hassle. Some researchers (e.g., Hsiao & Chen, 2012; Nageswaran et al., 2020; Su, 2009; Yalabik et al., 2005) show that MBGs are too generous to be optimal, and the partial-refund policies are more beneficial for retailers. Su (2009) studies the impact of product return policies on supply chain performance. Another format of partial-refund policy is implemented by charging a restocking fee (Shulman et al., 2009).

However, Shang et al. (2017) empirically discover that even the forward shipping fee decreases the value of the MBG return policy. Fruchter and Gerstner (1999) and Yalabik et al. (2005) find that offering a more generous return policy with free return shipping in addition to MBG can be optimal.

Our paper differs from previous literature in three main aspects. First, we are more interested in the impacts of reducing consumers' return shipping fees on a retailer's profit rather than designing the optimal refund amount for product returns. Our results add to prior work (e.g., Davis et al., 1998; Yalabik et al., 2005) by demonstrating how retailers can benefit from lowering the consumers' return shipping cost. Second, we introduce RSI, an innovative insurance-based return management tool, to reduce the consumers' return shipping costs. We find that RSI further improves retail profitability for any product return policies currently in use. Third, we capture the interplay among the insurer, retailer, and consumers in our model and illustrate that offering RSI can be a "win-win-win" proposition for them. Thus, our main contribution to product returns literature is characterizing how RSI can complement an existing return policy to benefit the retailer as well as consumers further.

Relevant literature on RSI is quite limited, as it is a recent innovation. Our work is most closely related to Zhang et al. (2022). They analyze the informational role of RSI when product quality is the retailer's private information and show that RRSI can be an effective signal of high quality. However, in practice, both the RRSI and CRSI policies are widely adopted in the online market. Our interests differ from Zhang et al. (2022) in that we investigate RSI from the perspective of product fit uncertainty. Consumers are heterogeneous in terms of their information-processing capabilities, and they may regret purchasing CRSI. Our results manifest that both the RRSI and CRSI policies may arise without quality uncertainty. In this regard, our work contributes to the prior research by theoretically supporting the viability of both RRSI and CRSI policies and providing practitioners with guidelines on adopting RSI. Recently, Chen et al. (2021) consider a manufacturer selling a product through a platform through agency selling or reselling format and investigate whether an RRSI strategy with an exogenously given premium should be offered. Apparently, the insurer plays an important role in a seller's decision to offer RSI in practice. Our work endogenizes the insurer's premium decision and characterizes the retailer's optimal RSI (RRSI or CRSI) policy. Li et al. (2021) attempt to investigate whether a retailer should charge RSI for a fee without modeling consumers' likelihood of product fit.

Our work is related to add-on pricing literature as RRSI and CRSI are essentially bundling and unbundling strategies of the product and insurance, respectively. Allon et al. (2011) explore whether a service provider should bundle a main service and an ancillary service, and unveil two motivations for unbundling the services, that is, altering consumer behavior or segmenting consumers. Fruchter et al. (2011) study a seller's strategy to offer a free add-on (i.e., bundling) or offer it for a fee (i.e., unbundling). They show that providing a free add-on can be more profitable for sellers with monopoly power. Shugan et al. (2017) consider different add-on bundling strategies with a product line, that is, a high- or low-end base. They find that the bundling strategies depend on the quality differentiation between base goods. Differently, Moon and Shugan (2018) focus on the bundle-framing effect and show that bundling can signal information about product popularity. Specifically, only sellers offering popular add-ons are incentivized to bundle add-ons with the main products. Recently, Cui et al. (2018) investigate the impact of the main service pricing, that is, uniform- or discriminatory-pricing, on the optimal add-on unbundling strategy. They find that a uniform-pricing (discriminatory-pricing) firm should unbundle the add-on if there is a large (small) fraction of high-type consumers who value the add-on. In addition, several papers extend the research on add-on pricing into competition settings (e.g., Ellison, 2005; Gabaix & Laibson, 2006; Geng & Shulman, 2015; Lin, 2017; Shulman & Geng,

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2013; Wang et al., 2019). Whereas all these papers study whether firms bundle or unbundle add-on, our paper studies add-on pricing in a distribution channel. Specifically, when the add-on RSI is unbundled (under CRSI policy), consumers purchase the RSI from the insurance company through the retailer's platform. Under the RRSI policy, the retailer purchases the RSI from the insurance company and sells the bundle of the RSI and the product. Moreover, we explore the effect of consumer-anticipated regret behaviors on retailers' RSI provision decisions.

Last, our work is related to the abundant literature on consumer-anticipated regret. Researchers in the operations management and marketing areas focus on the impact of the anticipated regret, including but not limited to, bid/auction (Engelbrecht-Wiggans & Katok, 2008; Filiz-Ozbay & Ozbay, 2007), product customization (Syam et al., 2008), advance selling (Diecidue et al., 2012; Nasiry & Popescu, 2012), product innovation (Jiang et al., 2017; Sarangee et al., 2013; Shih & Schau, 2011), product-line design (Zou et al., 2020), and workers' relocation decisions in an on-demand platform (Jiang et al., 2021). In particular, Jiang et al. (2017) show that anticipated regret has a nonmonotonic effect on firms' profits and the entrant's optimal quality; Zou et al. (2020) find that overpurchase regret hurts the firm but underpurchase regret can benefit the firm; and Jiang et al. (2021) show that regret-averse workers are more likely to relocate to the supply-shortage zone than rational workers. However, this does not necessarily benefit the firm. Also, Braun & Muermann (2004) utilize the concept of anticipated regret to explain consumers' desire for different kinds of insurance. Their model shows that regretful consumers adjust away from two extremes: full and no insurance. Inman & Zeelenberg (2002) also show that consumers may suffer from different levels of regret contingent on the cause of the regret. Recently, Li and Liu (2022) investigate the pricing policy of a regretful seller. They find that the seller sets a lower price as a result of the regret, but the pricing policy in terms of inventory and time is similar to those of unbiased sellers. Moreover, selling more goods may yield less revenue. Jin et al. (2022) explore how consumers' optimal search behavior is affected by the anticipated regret under a classic framework of sequential search. Their results show that the anticipated regret can intensify or soften price competition among firms.

Our paper investigates the impact of the consumers' anticipated regret on the RSI policy. Similar to the previous research, we use a linear structure to capture the anticipated regret in consumers' utility function and model two distinctive consumer regrets, namely, aversion to insured or uninsured regret. Our study enriches this research stream by identifying the salient effect of regret on RSI policies. Specifically, we explore the impact of regret when consumers are uncertain about the product fit. Moreover, we extend this research stream by incorporating the insurer's decision into the model, which has not been well addressed by prior literature.

3 | MODEL

Consider a monopolistic online retailer selling a product at price p. Throughout this paper, we use pronouns "he" for the retailer and "she" for the generic consumer. Consumers have different information-processing capabilities in inferring benefits from product attribute information due to different levels of expertise (Alba & Hutchinson, 1987). As in Jiang et al. (2017) and Sun and Gilbert (2019), we consider the following two consumer segments. A proportion θ of consumers know their true evaluation of the product at the time of purchase. We refer to these consumers as the *informed* consumers. Their willingness to pay for the product v is heterogeneous and uniformly distributed over $[0, 1]$.[8] The rest of the consumers $(1-\theta)$ are *uninformed*; they do not know, prior to purchasing, whether the product will satisfy their needs. However, they do know that with a probability α, the product is a perfect fit, and their willingness to pay for the product is $v \sim U[0, 1]$ as for the informed consumers. With probability $1-\alpha$, the product is not fit, failing to meet the consumers' needs; thus, these consumers' ex-post valuation of the product is zero (Huang et al., 2018; McWilliams, 2012). Therefore, uninformed consumers only ex post learn their true valuation for the product, equal to v with a probability α or zero with probability $1-\alpha$. Upon receiving misfit products, consumers can return these items to the retailer. Note that informed consumers would never need to do so as they know ex-ante their true valuation for the product. We assume that the probability of fit α is independent of the proportion of informed consumers θ as in Jiang et al. (2017) and confirm that the results qualitatively hold even if these two parameters are correlated.

The retailer adopts a full-refund return policy but does not cover the return shipping fee for consumers (e.g., 6pm.com[9]), so consumers must pay a unit shipping fee s for returns.[10] For every returned product, the retailer incurs a unit return handling cost h (including but not limited to repackaging the returned product to resell it).[11] We also investigate the partial-refund return policy by considering a restocking fee that consumers must pay to the retailer for product returns in the model extension (Section 6.1). Without loss of generality, we normalize the retailer's other operation costs, such as procurement cost, to zero and the market size to one.

The retailer decides on the product price p, as well as whether (1) to purchase the insurance at a premium λ_r for each product from an insurer to give RRSI to consumers for free,[12] or (2) to provide an option for consumers to purchase CRSI at an identical premium λ_c from the insurer directly.[13] Both premiums are determined by the insurer.[14] Having either RSI, consumers will be partially compensated by the insurer for their shipping cost associated with product returns. The compensation amount c is prespecified and is at most the shipping fee s actually incurred, that is, $c \le s$, which is consistent with the practice (refer to the CRSI [Figures A.1 and A.2][15] and RRSI [Figures A.3–A.6 in the

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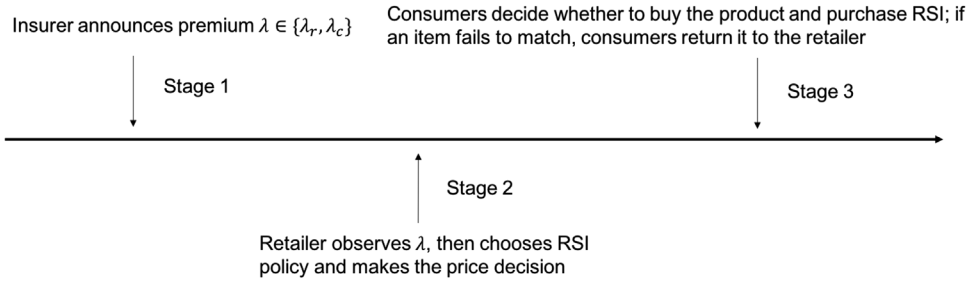


FIGURE 2 Game sequence.

Supporting Information][16] contracts on Taobao.com/Tmall.com in Appendix EC.2 in the Supporting Information). We suppose that the compensation amount is exogenously given because "in practice, the insurance compensation is a pre-specified amount calculated based on the route from a consumer's home to a retailer's return-handling facility, subject to a maximum amount per transaction" and "this reflects the fact that in practice the insurer usually has very limited flexibility in setting the compensation level" (Zhang et al., 2022). We consider a sufficiently low compensation amount ($c < \bar{c}$) to ensure the insurer's participation; otherwise, the results will be trivial. From the RSI contract between Taobao.com and ZhongAn Insurance (Figures A.1 and A.2 in the Supporting Information) or Huatai Insurance Group (Figures A.5 and A.6 in the Supporting Information), we see that the compensation amount is usually lower than the actual return shipping fee that a consumer pays to the delivery services company. For example, if it costs a consumer 20 RMB to ship a misfit product from Shanghai to Beijing, the insurer will reimburse 12 RMB of the shipping fee. Note that, at the time of purchase, the uninformed consumers are uncertain about the product fit and their need for RSI. Thus, under the CRSI policy, they may regret buying or not buying RSI, if they later find that the forgone alternative would have been a better choice than their RSI purchase decision (Braun & Muermann, 2004). To isolate the effect of consumer regret on RSI, we do not consider consumers' possible regret about product purchase decisions in the main model. In Section EC.1.2 in the Supporting Information, we show that the analysis can be performed in the same vein but the two regret effects are intertwined when the regret about product purchase is also incorporated.

The three-stage game sequence is shown in Figure 2. As pointed out by Zhang et al. (2022), the retailer and the uninformed consumers can observe the premiums before they make the RSI adoption decisions, so the insurer moves first to set the premiums, that is, the insurer announces premium $\lambda \in \{\lambda_r, \lambda_c\}$ in stage 1. In stage 2, the retailer chooses an RSI policy and makes the price decision after observing λ. In stage 3, consumers arrive and decide whether to buy the product and purchase RSI. If an item fails to match, consumers return it to the retailer. All the information is public. Table A.1 in the Supporting Information summarizes the notation. All the parameter thresholds and proofs are relegated to Appendix EC.3 in the Supporting Information.

4 | NO ANTICIPATED REGRET

We investigate the case with no consumer-anticipated regret in this section and will characterize the impact of consumer regret behaviors on the RSI policy in Section 5.

4.1 | Benchmark: In the absence of RSI

With no RSI available, the informed consumers' utility is $U_b^{\theta} = v - p_b$, and the uninformed consumers' utility is $U_b^{\bar{\theta}} = \alpha(v - p_b) - (1-\alpha)s$, where the subscript b denotes the benchmark case and the superscript $\bar{\theta}$ indicates the uninformed consumers. Therefore, the informed consumers with a valuation greater than the price p_b purchase the product, and the demand is $D_b^{\theta} = \theta(1 - p_b)$. The uninformed consumers take into consideration the possibility of the product being unfit, and thus the demand is $D_b^{\bar{\theta}} = (1-\theta)[1 - \frac{(1-\alpha)s}{\alpha} - p_b]$. The retailer's objective is to find the optimal product price p_b such that its profit $\Pi_{Rb} = D_b^{\theta} p_b + D_b^{\bar{\theta}}[\alpha p_b - (1-\alpha)h]$ is maximized. If the retailer's handling cost or consumers' return shipping cost is sufficiently high ($h \geq \bar{h}$ or $s \geq \bar{s}$), it would be more profitable for the retailer to sell products to the informed consumers only, who never return products, and the analysis will be trivial. Therefore, we assume $h < \bar{h}$ and $s < \bar{s}$ to avoid the trivial case throughout the paper. The benchmark equilibrium is summarized in Lemma 1.

Lemma 1. *In the benchmark case without RSI, the retailer sells to both the informed and uninformed consumers and the equilibrium price and profit are* $p_b^* = \frac{1}{2} - \frac{(1-\theta)(1-\alpha)(s-h)}{2(\alpha+\theta(1-\alpha))}$ *and* $\Pi_{Rb}^* = \Pi_{Rb}(p_b^*)$, *respectively.*

Next, we examine how RSI reduces the consumers' return shipping cost and benefits the retailer. Using backward induction, we first analyze the retailer's decision given an insurance premium in Section 4.2, and then characterize the insurer's optimal premium decision in Section 4.3.

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4.2 | Retailer's decision with given insurance premiums

In the presence of RSI,with a given premium $\lambda \in \{\lambda_r, \lambda_c\}$, the retailer decides whether to buy RSI and offer it to consumers for free (i.e., the RRSI policy; subscript r), or provide consumers the option of purchasing RSI themselves from the insurer (i.e., the CRSI policy; subscript c).

Suppose the retailer pays the premium λ_r, that is, adopting the RRSI policy, so uninformed consumers can freely benefit from a reduced shipping fee $s-c$ for product returns. Consequently, the utilities of the informed and uninformed consumers are $U_r^{\theta} = v - p_r$ and $U_r^{\bar{\theta}} = \alpha(v-p_r) - (1-\alpha)(s-c)$, respectively. Therefore, the demand of the informed consumers is $D_r^{\theta} = \theta(1-p_r)$, and the uninformed consumers' demand is $D_r^{\bar{\theta}} = (1-\theta)[1 - \frac{(1-\alpha)(s-c)}{\alpha} - p_r]$. Given the insurer's premium decision λ_r, the retailer sets the product price p_r to maximize its profit $\Pi_{Rr} = D_r^{\theta}(p_r - \lambda_r) + D_r^{\bar{\theta}}[\alpha p_r - (1-\alpha)h - \lambda_r]$. It is worth noting that the retailer does not know which consumers are informed, thus he has to pay the insurance premium λ_r for all consumers.

Under the CRSI policy, the informed consumers' utility is $U_c^{\theta} = v - p_c$ as they never purchase RSI; the corresponding demand is $D_c^{\theta} = \theta(1-p_c)$. The utilities of uninformed consumers who purchase RSI from the insurer at a premium λ_c and who do not purchase RSI are $U_{c1}^{\bar{\theta}} = \alpha(v-p_c) - (1-\alpha)(s-c) - \lambda_c$ and $U_{c0}^{\bar{\theta}} = \alpha(v-p_c) - (1-\alpha)s$, respectively. The uninformed consumers buy products with RSI if and only if this option gives them a higher utility, that is, $U_{c1}^{\bar{\theta}} > \max(0, U_{c0}^{\bar{\theta}})$, equivalently, $\lambda_c < (1-\alpha)c$. Correspondingly, their demand for the product as well as the insurance is $D_c^{\bar{\theta}} = (1-\theta)[1 - \frac{\lambda_c + (1-\alpha)(s-c)}{\alpha} - p_c]$. Given the insurer's premium decision λ_c, the retailer maximizes his profit $\Pi_{Rc} = D_c^{\theta} p_c + D_c^{\bar{\theta}}[\alpha p_c - (1-\alpha)h]$ by choosing the optimal product price p_c. Note that, each uninformed consumer either purchases both CRSI and the product, or makes no purchase. It is not this case anymore in the presence of consumer regret, as shown in Section 5; uninformed consumers with regret also may purchase just the product.

The equilibrium prices $p_r^*(\lambda_r)$ and $p_c^*(\lambda_c)$ are presented in Appendix EC.3 in the Supporting Information that includes other equilibrium prices derived in later analysis. Lemma 2 summarizes the retailer's optimal RSI adoption decision given the insurance premium $\lambda \in \{\lambda_r, \lambda_c\}$. Note: the superscript N denotes the no-regret case.

Lemma 2. *Given the insurance premium $\lambda \in \{\lambda_r, \lambda_c\}$, the retailer's optimal RSI policy and product price are as follows (see Figure A.7 in the Supporting Information).*

(*i*) *If the RRSI premium is relatively low ($\lambda_r < \tilde{\lambda}_r^N$), the retailer purchases the insurance for his consumers, that is, offers RRSI.*

(*ii*) *Otherwise, the retailer does not offer RRSI but adopts the CRSI policy only if the CRSI premium is relatively low ($\lambda_c < \tilde{\lambda}_c^N$).*

(*iii*) *The retailer always sets a higher product price under either RSI policy than in the benchmark case.*

Under the RRSI policy, as the retailer purchases RSI from the insurer, consumers can benefit from the insurance for free; this insurance policy alleviates the uninformed consumers' concerns about product returns. Accordingly, the retailer would set a higher product price than under the benchmark case so as to transfer the insurance premium cost to consumers and extract the uninformed consumers' elevated surplus stemming from the insurance benefit. With a higher product price, informed consumers are less likely to buy the product. For the uninformed consumers, RRSI boosts their utility by $(1-\alpha)c$ but the increased product price lowers their utility by $\alpha(p_r - p_b)$. Therefore, the uninformed consumers are more likely to purchase the product only when the insurance premium is not too high, that is, $\lambda_r < \bar{\lambda}_{r1}$, such that the product price goes up mildly. Further, the overall demand expands only when the premium is sufficiently low, that is, $\lambda_r < \bar{\lambda}_{r2}$ ($< \bar{\lambda}_{r1}$) and the uninformed consumers' demand expansion (DE) dominates the informed consumers' demand shrinkage (DS). Essentially, the RRSI policy repels some informed consumers who are sure of their needs and would never return products, but the same policy may attract more uninformed consumers who possibly return misfit products. Hence, this insurance policy exacerbates the adverse selection problem of the retailer, which we refer to as the *adverse selection exacerbation* (*AE*) effect. For a sufficiently small insurance premium ($\lambda_r < \bar{\lambda}_{r2}$), alleviated concerns for product returns expand the total market, referred to as the *DE* effect. When $\lambda_r > \bar{\lambda}_{r2}$, the overall demand shrinks, which we refer to as the *DS* effect.

A comparison of the retailer's profit under RRSI, CRSI, and the benchmark indicates that the retailer should offer RRSI when $\lambda_r < \tilde{\lambda}_r^N$. When $\lambda_r < \min(\bar{\lambda}_{r2}, \tilde{\lambda}_r^N)$, the AE and DE effects are countervailing forces for the interest of the retailer, and the RRSI policy favors the retailer only when the DE effect dominates the AE effect. In this case, the retailer employs a *volume-driven* strategy as the total demand expands. In contrast, if $\bar{\lambda}_{r2} < \lambda_r < \tilde{\lambda}_r^N$, the negative DS effect emerges, and both AE and DS effects hurt the retailer. However, the product price goes up drastically such that the retailer still profits more. In other words, the retailer adopts the *margin-driven* strategy. Interestingly, under the RRSI policy, the retailer makes full use of the value-added return insurance policy by setting a high "bundle price," that is, $p_r > p_b + \lambda_r$. As a result, the retailer not only transfers the insurance premium to consumers but also extracts their extra surplus.

When the insurer sets a high RRSI premium ($\lambda_r > \tilde{\lambda}_r^N$) and low CRSI premium ($\lambda_c < \tilde{\lambda}_c^N$), offering consumers a choice to purchase CRSI from the insurance company benefits the retailer more than RRSI. A low CRSI premium raises the uninformed consumers' willingness to pay for the

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product because the expected return shipping cost is reduced by c. Therefore, the retailer would charge a higher product price than in the benchmark case. Accordingly, fewer informed consumers make purchases, and the AE effect comes into play. CRSI benefits the retailer only when the premium is so low that more uninformed consumers will purchase and the overall demand still expands such that the DE effect outweighs the AE effect. However, we next show that CRSI is not viable when we take into consideration the insurer's profit.

4.3 | Optimal insurance premiums

The insurer charges the premiums λ_r and λ_c under RRSI and CRSI, respectively, but expects to compensate c to any uninformed consumer with a probability $1-\alpha$. Under RRSI, the insurer decides on the optimal RRSI premium λ_r by maximizing its profit $\Pi_{Ir} = D_r^{\theta}\lambda_r + D_r^{\bar{\theta}}[\lambda_r - (1-\alpha)c]$. Under the CRSI policy, the insurer decides on the optimal CRSI premium λ_c by maximizing its own benefit $\Pi_{Ic} = D_c^{\bar{\theta}}[\lambda_c - (1-\alpha)c]$. Note that, when the insurer sets the insurance premiums at the boundary value ($\lambda_r = \tilde{\lambda}_r^N$ and $\lambda_c = \tilde{\lambda}_c^N$), the retailer would not have the incentive to adopt the insurance since the profit remains unchanged. Therefore, we consider only the interior solutions in the following analysis. Moreover, we must have $\lambda_c > (1-\alpha)c$ such that the insurer is incentivized to participate in offering CRSI. Meanwhile, this constraint would lead to a premium that is so high that the CRSI is no longer attractive to consumers. As summarized in Proposition 1, the CRSI is not viable for any endogenous insurance premium.

Proposition 1. *In equilibrium,*

(*i*) *if the retailer's return handling cost is relatively low* ($h_{r1}^N < h < h_{r2}^N$) *and consumers' return shipping cost is medium* ($s_{r1}^N < s < s_{r2}^N$), *the insurer sets premiums such that the retailer would offer RRSI and charge a higher product price than in the benchmark case;*
(*ii*) *otherwise, the insurer does not offer RSI.*

As indicated in Lemma 2, under RRSI, the retailer sets a premium "bundle" product price to not only transfer the insurance cost to consumers but also absorb the uninformed consumers' increased surplus. With a higher product price, fewer informed consumers make a purchase, and AE is always in effect. The retailer employs volume-driven and margin-driven strategies for relatively small and high premiums set by the insurer, respectively. In the volume-driven mode, the number of informed consumers leaving the market is less than the number of uninformed consumers entering the market. This happens only when the consumers' return shipping cost (s) is relatively high ($\max(s_{r1}^N, \bar{s}_r) < s < s_{r2}^N$) and the retailer does not drive the product price so high; under those conditions, consumers greatly enjoy the extra insurance benefit. For the retailer, offering the RRSI policy under this volume-driven strategy is a double-edged sword: On the one hand, more consumers purchase the product (DE effect); on the other hand, more products will be returned, and fewer informed consumers will be served (AE effect). For the RRSI to be profitable, the retailer's return handling cost must be relatively small ($h < h_{r2}^N$) to limit the negative effect from AE.

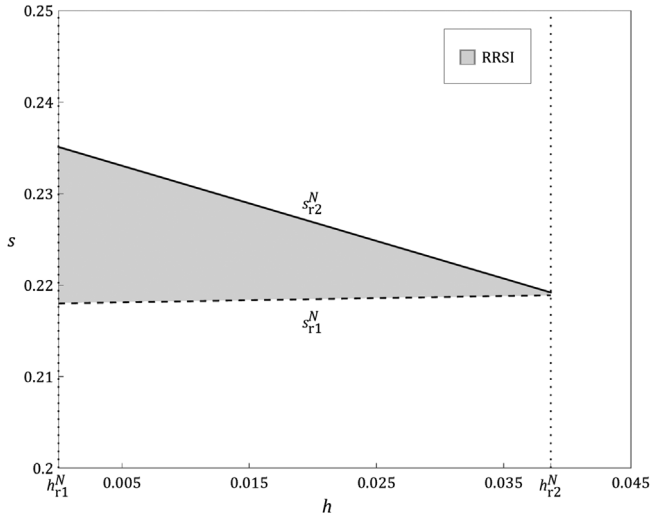


FIGURE 3 Optimal return shipping insurance (RSI) policy without consumers' anticipated regret. *Note*: $\theta = 0.05, \alpha = 0.25, c = 0.05$.

If the return shipping cost (s) is relatively low ($s_{r1}^N < s < \bar{s}_r$), then the insurer can set a higher premium, and the retailer raises the product price drastically. Then, the margin-driven strategy is more beneficial. In this setting, both AE and DS have negative effects on the retailer's profitability, but the premium "bundling" pricing strategy improves the profit margin. Moreover, the negative DS effect is limited only if the retailer's return handling cost exceeds a threshold ($h > h_{r1}^N$). As such, the RRSI still benefits the retailer, but it applies only in an interval of return handling cost. In sum, as illustrated in Figure 3, the RRSI policy is optimal when $h_{r1}^N < h < h_{r2}^N$ and $s_{r1}^N < s < s_{r2}^N$.

Zhang et al. (2022) show that the RRSI policy is never optimal if consumers have no quality uncertainty regarding the product. However, in practice, RRSI is commonly used for goods like televisions and mobile phones. For these products, consumers can easily acquire product quality information by searching online. Unlike their work, we address the RSI policy from the perspective of product fit uncertainty, and we segment consumers according to their information-processing capabilities: informed and uninformed. From this new angle, our results complement the literature by considering fit uncertainty and potentially explaining the prevalence of RRSI in e-commerce markets.

Nevertheless, Proposition 1 shows that the insurer should never offer CRSI, which is inconsistent with real-world practice. Resolving this inconsistency was the motivation for incorporating the consumers' anticipated regret behaviors in our analysis in Section 5.

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5 | ANALYSIS WITH ANTICIPATED REGRET

Now we explore the optimal RSI policy when consumers anticipate potential post–purchase regret about their RSI purchase decisions. Informed consumers will never experience regret ex-post because they know their true valuation for the product before purchasing, and thus they never need to return products. In contrast, uninformed consumers are uncertain about the product fit and their need for RSI during purchasing. They will experience regret if they find that the forgone alternative would have been a better choice than their RSI purchase decision (Braun & Muermann, 2004). For instance, if an uninformed consumer purchases the product without buying RSI but later has to return the product due to misfit, she will regret not having bought the RSI, from which she would have derived a higher utility ex-post; we refer to this as *uninsured regret*. Similarly, the consumer will experience regret for buying the product with RSI if she receives a product that fits well; we refer to this as *insured regret*. Note that consumers may experience regret after purchasing even though their product purchase decisions are optimal. Our survey shows that 80.00% of the respondents anticipate insured regret and 96.65% of the respondents anticipate uninsured regret. Therefore, the uninformed consumers' purchase decisions are based on their expected utility taking into account the disutility from the anticipated potential post–purchase regret regarding RSI (Zeelenberg et al., 1996).

To incorporate the anticipated regret in the consumer utility function, we use a linear regret term similar to the technique in Syam et al. (2008) and Jiang et al. (2017). Specifically, the disutility induced by regret aversion is proportional to the utility difference between the forgone and chosen alternatives (Inman et al., 1997). Therefore, the anticipated regret results in a disutility equal to $-\gamma \text{Prob}(U_f > U_c)(U_f - U_c)$, where γ is consumer's aversion sensitivity toward regret, U_f and U_c are the utility of forgone and chosen alternatives, respectively. Prior studies (Inman & Zeelenberg, 2002; Jiang et al., 2017; Nasiry & Popescu, 2012) also show that consumers have different levels of regret contingent on the cause of the regret. In our setting, we use γ_I and $\gamma_N^i (i \in \{H, L\})$ to denote the consumer's aversion sensitivity to the insured and uninsured regret, respectively. Moreover, consumers differ in their propensity for anticipated regret. Here, we assume consumers are heterogeneous in their sensitivity to uninsured regret: A $\beta (0 < \beta < 1)$ portion of the uninformed consumers have low sensitivity to the uninsured regret, that is, $\gamma_N^i = \gamma_N^L$ (i.e., L denotes uninformed consumers with low uninsured regret); the rest of $1 - \beta$ uninformed consumers have high sensitivity to the uninsured regret, that is, $\gamma_N^i = \gamma_N^H > \gamma_N^L$ (i.e., H denotes uninformed consumers with high uninsured regret). Alternatively, we can assume that consumers are heterogeneous in their sensitivity to the insured regret, which will lead to similar results.

Using backward induction, we first analyze the retailer's decision given the insurance premiums in Section 5.1, and then characterize the insurer's optimal premium decision in Section 5.2. Based on the equilibrium analysis, we examine the impacts of consumers' regret behaviors on equilibrium outcomes and the impact of RSI policies on stakeholders in Sections 5.3 and 5.4, respectively.

5.1 | Retailer's decision with given insurance premiums

With the anticipated regret, under CRSI, uninformed consumers will experience insured or uninsured regret ex-post, if their RSI purchase decisions result in a lower utility than the forgone alternative. In particular, uninformed consumers derive a utility of $U_{c1}^{\bar{\theta}} = \alpha(v - p_c - \gamma_I \lambda_c) - (1 - \alpha)(s - c) - \lambda_c$, where $\gamma_I \lambda_c$ is the disutility of the insured regret, for the CRSI purchasing option. If uninformed consumers do not purchase CRSI, the utility is $U_{c0}^{\bar{\theta}} = \alpha(v - p_c) - (1 - \alpha)[s + \gamma_N^i (c - \lambda_c)]$, where $\gamma_N^i (c - \lambda_c)$ is the disutility from the uninsured regret. The uninformed consumers will buy the product with CRSI if and only if this option gives them a higher utility, that is, $U_{c1}^{\bar{\theta}} > \max(0, U_{c0}^{\bar{\theta}})$; this situation occurs when the insurance premium is not too expensive, that is, $\lambda_c < \bar{\lambda}_c^R(\gamma_N^i) = \frac{(1+\gamma_N^i)(1-\alpha)c}{1+\alpha\gamma_I+(1-\alpha)\gamma_N^i}$. Note that $(1 - \alpha)c < \bar{\lambda}_c^R(\gamma_N^i) < c$.

Specifically, if the CRSI has a relatively high premium, that is, $\bar{\lambda}_c^R(\gamma_N^L) < \lambda_c \le \bar{\lambda}_c^R(\gamma_N^H)$, only H uninformed consumers consider buying RSI while L uninformed consumers are interested in the product only (Case 1). In Case 1, the demands from H and L uninformed consumers are $D_{c1}^{\bar{\theta}H} = (1 - \theta)(1 - \beta)[1 - \frac{(1+\alpha\gamma_I)\lambda_c+(1-\alpha)(s-c)}{\alpha} - p_c]$ and $D_{c1}^{\bar{\theta}L} = (1 - \theta)\beta[1 - \frac{(1-\alpha)(\gamma_N^L(c-\lambda_c)+s)}{\alpha} - p_c]$, respectively. When the CRSI premium is sufficiently low, that is, $\lambda_c \le \bar{\lambda}_c^R(\gamma_N^L)$, both H and L uninformed consumers consider buying the product with RSI (Case 2). In Case 2, the demands from H and L consumers are $D_{c2}^{\bar{\theta}H} = (1 - \theta)(1 - \beta)[1 - \frac{(1+\alpha\gamma_I)\lambda_c+(1-\alpha)(s-c)}{\alpha} - p_c]$ and $D_{c2}^{\bar{\theta}L} = (1 - \theta)\beta[1 - \frac{(1+\alpha\gamma_I)\lambda_c+(1-\alpha)(s-c)}{\alpha} - p_c]$, respectively. We use $K \in \{1, 2\}$ to indicate the CRSI cases and thus the demand of the uninformed consumers is $D_{cK}^{\bar{\theta}} = D_{cK}^{\bar{\theta}H} + D_{cK}^{\bar{\theta}L}$. Given the insurer's premium decision λ_c, the retailer maximizes its profit $\Pi_{Rc} = D_c^{\theta} p_c + D_{cK}^{\bar{\theta}}[\alpha p_c - (1 - \alpha)h]$ by choosing the optimal product price p_c. We summarize the retailer's optimal RSI adoption decision given the insurance premium in Lemma 3. Note that the superscript R denotes the case with regret.

Lemma 3. *Given the insurance premium $\lambda \in \{\lambda_r, \lambda_c\}$, the retailer's optimal RSI policy and product price are as follows (see Figure A.8 in the Supporting Information).*

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(*i*) *If the RRSI premium is relatively low* ($\lambda_r < \tilde{\lambda}_r^R$), *the retailer purchases the insurance for the consumers, that is, offers RRSI.*

(*ii*) *Otherwise,*

(*a*) *the retailer offers CRSI and sets the product price such that both H and L uninformed consumers purchase CRSI* (*Case* 2) *if* $\lambda_c < \tilde{\lambda}_{c2}^{R\dagger}$ *or* $\tilde{\lambda}_{c2}^{R\ddagger} < \lambda_c \le \bar{\lambda}_c^R(\gamma_N^L)$;

(*b*) *the retailer offers CRSI and sets the product price such that only H uninformed consumers purchase CRSI* (*Case* 1) *if* $\bar{\lambda}_c^R(\gamma_N^L) < \lambda_c < \tilde{\lambda}_{c1}^{R\dagger}$ *for* $\beta > \tilde{\beta}_c$ *or* $\tilde{\lambda}_{c1}^{R\ddagger} < \lambda_c \le \bar{\lambda}_c^R(\gamma_N^H)$ *for* $\beta \le \tilde{\beta}_c$.

(*iii*) *Compared with the benchmark case, the retailer always sets a higher product price under RRSI, but under CRSI, he sets a higher product price if and only if the CRSI has an extremely low premium, that is,* $\lambda_c < \tilde{\lambda}_{c2}^{R\dagger}$.

Under RRSI, the consumers' anticipated regret does not play a role in their purchasing decisions, and thus the insights are the same as in the case without regret (Lemma 2). The retailer should offer RRSI only when the premium is sufficiently low ($\lambda_r < \tilde{\lambda}_r^R$); otherwise, CRSI should be considered. Now uninformed consumers who purchase CRSI incur an insured regret disutility of $\gamma_I \lambda_c$ with probability α; and those not purchasing CRSI experience an uninsured regret disutility of $\gamma_N^i(c - \lambda_c)$, with probability $1 - \alpha$. Both H and L uninformed consumers consider the CRSI option when the CRSI premium is sufficiently low ($\lambda_c < \bar{\lambda}_c(\gamma_N^L)$), and only H uninformed consumers consider CRSI option for intermediate premiums ($\bar{\lambda}_c(\gamma_N^L) < \lambda_c < \bar{\lambda}_c(\gamma_N^H)$). Otherwise, CRSI is not attractive to any consumer.

On the one hand, CRSI alleviates uninformed consumers' concerns about product misfits. On the other hand, consumers will need to pay for the insurance service and make cost–benefit trade-off decisions. Interestingly, an extremely low premium ($\lambda_c < \tilde{\lambda}_{c2}^{R\dagger}$) enables the retailer to exploit the consumers' extra value that comes from the insurance benefit and charge a higher product price than in the benchmark; otherwise, charging a lower product price is necessary to attract more uninformed consumers. As a result, more informed consumers purchase the product, and the *adverse selection mitigation* (AM) effect emerges for a high premium, and the opposite AE effect exists otherwise. Moreover, the insured (uninsured) regret is more pronounced for a high (low) premium, making uninformed consumers less (more) likely to purchase CRSI.

Consequently, for an extremely small premium ($\lambda_c < \tilde{\lambda}_{c2}^{R\dagger}$), the high uninsured regret brings more demand. Furthermore, the DE effect dominates the AE effect and thus offering CRSI benefits the retailer. When $\lambda_c > \tilde{\lambda}_{c2}^{R\dagger}$, the AM effect emerges. However, the high insured regret plays a more important role in the consumers' purchasing decision, so fewer consumers purchase, that is, the DS effect arises. The AM and DS effects are in conflict for the retailer's interest, so CRSI policy favors the retailer only when the AM effect dominates the DS effect,

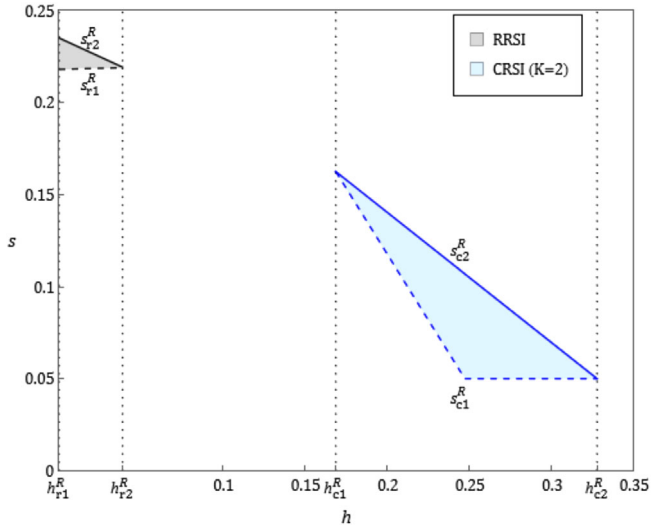


FIGURE 4 Optimal return shipping insurance (RSI) policy with consumers' anticipated regret. *Note*: $\theta = 0.05, \alpha = 0.25, c = 0.05, \beta = 0.4, \gamma_N^H = 0.8, \gamma_N^L = 0.4, \gamma_I = 0.2$.
[Color figure can be viewed at wileyonlinelibrary.com]

that is, $\tilde{\lambda}_{c2}^{R\ddagger} < \lambda_c < \bar{\lambda}_c^R(\gamma_N^L)$, $\bar{\lambda}_c^R(\gamma_N^L) \le \lambda_c < \tilde{\lambda}_{c1}^{R\dagger}$ for $\beta > \tilde{\beta}_c$, or $\tilde{\lambda}_{c1}^{R\ddagger} < \lambda_c < \bar{\lambda}_c^R(\gamma_N^H)$ for $\beta \le \tilde{\beta}_c$. However, we next show that CRSI is actually not viable when $\lambda_c < \tilde{\lambda}_{c2}^{R\dagger}$, taking into consideration the insurer's profit.

5.2 | Optimal insurance premiums

Now we examine the insurer's optimal premium decisions (λ_r and λ_c) by maximizing its profits $\Pi_{Ir} = D_r^{\theta}\lambda_r + D_r^{\bar{\theta}}[\lambda_r - (1-\alpha)c]$ and $\Pi_{Ic} = [D_{cK}^{\bar{\theta}H} + (K-1)D_{cK}^{\bar{\theta}L}][\lambda_c - (1-\alpha)c]$, respectively. Note that $\lambda_c > (1-\alpha)c$ is necessary to ensure the insurer's participation incentives under CRSI. Proposition 2 and Figure 4[17] show that CRSI becomes viable in the presence of consumer regret.

Proposition 2. *In equilibrium,*

(*i*) *if the retailer's return handling cost is relatively low* ($h_{r1}^R < h < h_{r2}^R$) *and the consumers' return shipping cost is medium* ($s_{r1}^R < s < s_{r2}^R$), *the insurer sets premiums such that the retailer offers RRSI and charges a higher product price than in the benchmark case;*

(*ii*) *if H uninformed consumers are more sensitive to the uninsured regret than the insured regret* ($\gamma_N^H > \gamma_I$), *the return handling cost of the retailer is relatively high* ($h_{c1}^R < h < h_{c2}^R$) (note $h_{c1}^R > h_{r2}^R$), *and the consumers' return shipping cost is medium* ($s_{c1}^R < s < s_{c2}^R$), *then the insurer sets premiums such that the retailer offers CRSI and charges a lower product price than in the benchmark case;*

(*iii*) *otherwise, the insurer never offers RSI.*

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The anticipated regret does not impact the RRSI policy, so the intuitions and insights are consistent with those discussed for Proposition 1. Lemma 3 shows that, for given insurance premiums, offering CRSI benefits the retailer when the corresponding premium is extremely small ($\lambda_c < \tilde{\lambda}_{c2}^{R\dagger}$). However, the insurer would not gain any profit from a too small premium, so $\lambda_c > (1-\alpha)c > \tilde{\lambda}_{c2}^{R\dagger}$ must be met. Therefore, only when the consumers' return shipping cost is in a medium range ($s_{c1}^R < s < s_{c2}^R$) are consumers willing to realize the insurance benefit at the expense of a high premium; in this case, the insurer would offer a profitable CRSI. Then, the CRSI policy entails two countervailing effects: Fewer consumers make a purchase (DS effect), and more informed consumers purchase the product (AM effect). The negative DS effect is limited if the selling price does not decline too much compared to the benchmark. As such, the positive AM effect outweighs the negative DS effect only when the handling cost is in a medium range ($h_{c1}^R < h < h_{c2}^R$). To summarize, the CRSI policy emerges in the equilibrium only when both the retailer's and consumers' return costs are within a range.

In general, as the insured regret level γ_I goes up, the overall consumer demand decreases (the negative DS effect), the informed consumers' demand increases (the positive AM effect) more significantly, and the net positive effect diminishes. As a matter of fact, if consumers are highly averse to the insured regret, that is, $\gamma_I \geq \gamma_N^H$, both the DS and AM effects disappear because no uninformed consumers are interested in CRSI due to the high premium. Then, the CRSI ceases to be profitable. Therefore, CRSI is not viable absent consumers' anticipated regret behavior. In other words, consumers' strong uninsured regret directs them to take advantage of CRSI, which is the fundamental driving force for the feasibility of CRSI. H uninformed consumers are more sensitive to the uninsured regret than L uninformed consumers, so CRSI can be attractive to consumers only when at least the H uninformed consumers' uninsured regret level is higher than the insured regret level ($\gamma_N^H > \gamma_I$). Indeed, our survey shows that 44.44% of the respondents are more sensitive to the uninsured regret than the insured regret.

Proposition 2 reveals that the retailer should offer RRSI when the return handling cost is relatively low. In practice, major retailers on Taobao.com and JD.com pay less than small retailers to repackage returned products and resell them, that is, their return handling cost is lower. Recall that in the GAP example shown in Figure 1, the GAP flagship store can handle returns efficiently by selling the returned items through its widespread offline outlets in addition to its official flagship store on Taobao.com. Conversely, small sellers incur a higher cost to resell returned products. This potentially explains why major retailers tend to adopt RRSI, while small sellers prefer CRSI in online markets.

Zhang et al. (2022) show that the CRSI policy never arises in equilibrium, as the insurer never finds it is profitable to sell this insurance to consumers. However, the CRSI policy is indeed adopted by some retailers in practice, such as by some small stores on Taobao.com (see the GAP example in Figure 1b). By considering the consumers' regret behaviors in our model, we uncover that both RRSI and CRSI can arise in equilibrium. Our results potentially explain different RSI policies prevalent in the e-commerce market.

It is worth noting that the positive AM effect of anticipated regret, which makes CRSI optimal, hinges on the market structure consisting of informed and uninformed customers and the assumption that the retailer cannot price discriminate between these two segments. With consumers' anticipated regret behavior, the retailer has to lower the selling price, which attracts more informed customers. In practice, if a retailer chooses to provide RSI (RRSI or CRSI), the RSI must be for all consumers, as shown in Figures A.4 and A.6 in the Supporting Information. Platforms may require sellers to offer MBG to all purchases. Only in such markets where retailers cannot separate out informed consumers can CRSI be an effective instrument to mitigate consumers' concerns about fit uncertainty.

From the theoretical perspective, our work complements the prior studies in product return and MBG. Retailers have long been offering generous return policies, but researchers still debate the optimality of MBGs (Abdulla et al., 2019). Huang et al. (2018) suggest that each firm should adopt MBGs regardless of any competitor's choice. Yet, consumers are still responsible for the return shipping fee, which is one salient factor in the consumers' purchasing decisions, especially for fashion industry goods such as apparel and beauty supplies. The survey conducted by Invesp indicates that 79% of consumers want free return shipping (Spillane, 2022), but this topic has been underexplored in the literature. We endeavor to fill this gap by examining the prevalent RSI policy. We unveil that RSI may improve the profitability of retailers who offer "MBG + RSI," a more generous and profitable policy than MBG. As a matter of fact, in Section 6.1, we demonstrate that RSI is a profitable instrument for a partial refund return policy as well.

Our work also adds to the literature on consumers' anticipated regret. Many researchers have explored the role of anticipated regret in different contexts, such as competition between standard and customized products (Syam et al., 2008), advance selling (Nasiry & Popescu, 2012), product innovation (Jiang et al., 2017), product-line design (Zou et al., 2020), on-demand service platforms (Jiang et al., 2021), and sequential consumers search framework (Jin et al., 2022). In particular, Jiang et al. (2017) discover that consumers' strong switching regret differentiates the uninformed consumers more from the informed ones in a duopoly market. The alleviated price competition encourages the entrant to invest more in innovation and gain more because consumers with regret behavior are willing to pay more for innovative products. Zou et al. (2020) show that consumers' stronger underpurchase regret than overpurchase regret stimulates consumers to consider the high-quality product. As a result, the firm can increase the quality gap in the product line when the underpurchase regret is strong.

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Differently, our paper focuses on consumers' uncertain preference for product fit rather than quality and explores how consumers' anticipated regret affects their purchase of the product and RSI. We unveil that consumers' stronger uninsured regret than the insured regret avails the optimality of CRSI. In addition, the insurer's decision is incorporated into our framework. Interestingly, consumers' regret behavior can lead to a "win-win-win" outcome for the insurer, retailer, and consumers as shown in Proposition 4 in Section 5.4.

Moreover, we point out two distinctive types of consumer regret, that is, insured regret and uninsured regret, that play important roles in the RSI policy. In the absence of regret, when consumers' willingness to buy products with CRSI is low, the insurer cannot set a profitable premium that consumers accept. As a result, the CRSI policy is never viable, which cannot explain the practice of commonly used CRSI policies. In the presence of regret, consumers' high uninsured regret relative to the insured regret enhances their willingness to pay for CRSI, which assures the insurer's profitability and enables CRSI to appear in equilibrium.

From the practice perspective, Proposition 2 highlights the crucial role of the consumers' high uninsured regret level in the retailer's RSI policy adoption decision. This suggests that retailers and insurers may advertise a cheap insurance premium and make the potential loss from paying the return shipping cost prominent to increase the consumers' uninsured regret level in order to induce more consumers to purchase CRSI. Indeed, retailers on Taobao.com highlight the compensation amount and the relatively cheap insurance premium to evoke the consumers' uninsured regret (see the GAP example in Figure 1b). In our survey, among respondents who feel the uninsured regret is stronger than the insured regret, most (67.21%) state that the main reason is the cheap insurance premium and relatively high compensation.

5.3 | Impact of the anticipated regret on equilibrium outcomes

We first investigate the impact of the anticipated regret on equilibrium outcomes.

Proposition 3. *The consumers' anticipated regret impacts the equilibrium outcomes as follows:*

(*i*) *The CRSI policy is viable only if the consumers' insured regret is relatively small* ($\gamma_I < \gamma_N^H$).

(*ii*) *As consumers become more sensitive to the insured regret* (γ_I *increases up to* γ_N^H),
  (*a*) *the insurer charges a lower premium, sells less CRSI, and profits less;*
  (*b*) *the retailer charges a lower product price, attracts*(*repels*) *more* (*un*)*informed consumers, and profits more.*

(*iii*) *As L uninformed consumers become more sensitive to the uninsured regret* (γ_N^L *increases up to* γ_I *such that only H uninformed consumers purchase the insurance*),
  (*a*) *the insurer sets a higher premium if and only if the consumers' return shipping cost is relatively high* ($s > \bar{s}_{c1}$), *but always sells more CRSI and profits more;*
  (*b*) *the retailer charges a higher product price and thus repels* (*attracts*) *more* (*un*)*informed consumers if and only if both L uninformed consumer segment is small and consumers' return shipping cost is low* ($\beta < \tilde{\beta}_c$ *and* $s < \bar{s}_{c2}$); *the retailer profits more if both the L uninformed consumer segment is small and the consumers' return shipping cost is low enough* ($\beta < \tilde{\beta}_c$ *and* $s < \bar{s}_{c3}$), *or the consumers' return shipping cost is sufficiently high* ($s > \bar{s}_{c4}$).

Proposition 3 highlights the importance of the consumers' anticipated regret behaviors under CRSI. As shown in Proposition 1, in the absence of regret, the insurer cannot offer a profitable CRSI that consumers desire. Moreover, the uninsured regret stimulates the consumers' longing for CRSI, while the insured regret has the opposite effect. Therefore, CRSI is attractive to consumers only when some consumers are more sensitive to uninsured regret, that is, $\gamma_I < \gamma_N^H$. In this range, fewer consumers intend to take the insurance benefit, and thus the insurer charges a lower premium and profits less as the insured regret aversion sensitivity γ_I goes up. Interestingly, with the increase of this sensitivity γ_I, the retailer would profit more! This is because the retailer can mitigate the adverse selection problem by offering a lower product price, which attracts more informed consumers who do not return products but repels more uninformed consumers.

One might conjecture that both the insurer and retailer would benefit from the consumers' increasing uninsured regret aversion sensitivity γ_N^i. Proposition 3 reveals that the interplay between this sensitivity and the consumers' return shipping cost is intricate. First, H uninformed consumers purchase the product with CRSI for medium product return costs when $\gamma_N^H > \gamma_I$ (Proposition 2). Accordingly, further increasing γ_N^H does not change the optimal RSI policy. Second, when $\gamma_N^L > \gamma_I$ (Case 2), the CRSI may attract L uninformed consumers as well. Therefore, only when $\gamma_N^L < \gamma_I$ (Case 1) will L uninformed consumers experience uninsured regret. Although H uninformed consumers purchase either both the product with CRSI or nothing, the change of γ_N^L still affects the purchasing behavior of H uninformed consumers indirectly through product price set by the retailer and insurance premium set by the insurer.

As γ_N^L increases up to γ_I, L uninformed consumers will experience uninsured regret more and be more likely to leave the market. At first glimpse, one would speculate that the insurer should not change the premium with γ_N^L as L uninformed consumers do not buy CRSI as long as $\gamma_N^L < \gamma_I$. Interestingly, we find that the insurer can charge a higher premium when the consumers' return shipping cost exceeds a threshold ($s > \bar{s}_{c1}$). This is because L uninformed consumers' uninsured regret (i.e., $\gamma_N^L(c - \lambda_c)$) declines with a higher insurance premium, which means that the retailer can

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retain revenues from L uninformed consumers coupled with a lower product price. It turns out that H uninformed consumers are more willing to make purchases as the overall effect of a higher premium and lower product price drives their actual cost down ($\frac{\partial[\lambda_c+\alpha(p_c+\gamma_I\lambda_c)+(1-\alpha)(s-c)]}{\partial\gamma_N^L} < 0$). When consumers' return shipping cost is relatively low, intuitively, the insurer lowers the premium to lower the insured regret of H uninformed consumers and thereby sell more. As a result, the insurer always sells more CRSI and profits more as L uninformed consumers are more sensitive to uninsured regret.

As the insurance premium λ_c changes, the retailer adjusts his price accordingly, contingent on the uninformed consumer segment structure. When sufficiently many H uninformed consumers are in the market ($\beta < \tilde{\beta}_c$) and the consumers' return shipping cost is low enough ($s < \bar{s}_{c2}(< \bar{s}_{c1})$) that the insurer sets a lower premium, the retailer prices the product higher because the overall effect of a lower premium and higher product price still yields a higher demand from H uninformed consumers. Further, the increasing uninsured regret actually benefits the retailer as both the product price and total demand from uninformed consumers go up when sufficiently many H uninformed consumers exist ($\beta < \tilde{\beta}_c$) and there is a low return shipping cost ($s < \bar{s}_{c3}(< \bar{s}_{c2})$). Moreover, if the return shipping cost is relatively high ($s > \bar{s}_{c4}(\geq \bar{s}_{c2})$), the retailer sets a lower product price to exploit the informed consumer segment by mitigating the adverse selection problem. Consequently, the retailer gains a higher profit, regardless of the uninformed consumer segment structure.

Proposition 3 reflects that the retailer and insurer should manipulate the consumer propensity for uninsured regret in their favor. For example, they can advertise the drawbacks of not paying a small premium for a high compensation benefit in case of returns. The reinforced uninsured regret level stimulates the consumers' desire for the CRSI policy, which benefits the retailer, insurer, and possibly consumers, as shown in Proposition 4 in Section 5.4.

We next explore the magnitude of the insurer's and the retailer's benefits from adopting the optimal RSI and how the regret sensitivity impacts these benefits. We focus on the impact of regret aversion sensitivities γ_I and γ_N^L on the benefits. The retailer's benefits are measured by the average profit improvement percentage in the regions s and h where RSI is optimal, that is, $\frac{\Pi_{Rr}-\Pi_{Rb}}{\Pi_{Rb}}$ or $\frac{\Pi_{Rc}-\Pi_{Rb}}{\Pi_{Rb}}$. Note that, the insurer only profits when RSI is adopted. To characterize how the regret sensitivity levels impact the insurance's benefit, we calculate the profit improvement relative to the lowest profit the insurer can gain, that is, $\frac{\Pi_{Ir}-min\Pi_{Ir}}{min\Pi_{Ir}}$ or $\frac{\Pi_{Ic}-min\Pi_{Ic}}{min\Pi_{Ic}}$.

For illustration, we focus on Case 2, where both H and L consumers consider the product with RSI, so we choose parameters such that $\gamma_I < \gamma_N^L < \gamma_N^H$. Specifically, with a step of 0.05, we vary γ_I from 0 to $\gamma_N^L - 0.05$ with $\gamma_N^L = 0.95$ to investigate the impact of γ_N^L on profits, and vary γ_N^L from $\gamma_I + 0.05$ to $\gamma_N^H - 0.05$ with $\gamma_I = 0.05$ to investigate the impact of γ_N^L on profits. In addition, we vary h from 0 to 0.35 and s from 0 to 0.25 with a step of 0.0025. In total, we generate 14,241 problem instances. We calculate the average profit improvement percentages across instances for each value of γ_I and γ_N^L, respectively.

Figure A.9 in the Supporting Information depicts how the average magnitude of the retailer's profit improvement changes with γ_I and γ_N^L, respectively, compared to the benchmark. The figure shows that RRSI improves the retailer's profit at an average rate above 1.7%. As γ_I and γ_N^L go up, the retailer's profit improvement from adopting CRSI increases with γ_I and γ_N^L and the average improvement rate is from 2.1% to 4.1% and from 0.5% to 2.2%, respectively. In other words, the average rate can reach up to 4.1%. The online retail sales were 13.1 trillion Chinese Yuan (CNY) and $959.9 billion in China and the United States, respectively, in 2021 (OBERLO, 2022; Xinhua, 2022). This demonstrates that RSI can benefit retailers significantly.

Figure A.10 in the Supporting Information shows how γ_I or γ_N^L impacts the insurer's average profit improvement compared to the lowest possible profit. We can see that the insurer's profit can increase up to 15 times when consumers' regret level changes. As of June 2022, ZhongAn Insurance Company earned 2.027 billion CNY from RSI premiums, an annual change of 38% (Tu & Liao, 2022). This suggests that insurers should manipulate consumers' regret propensity in their favor.

5.4 | Impact of the RSI

The adoption of RSI policies at equilibrium certainly benefits both the retailer and the insurer. Proposition 4 examines whether consumers are better off as well.

Proposition 4. *Consumers are worse off with RRSI, while CRSI leads to a "win-win-win" outcome for the retailer, insurer, and consumers when* $h_{c1}^R < h < \bar{h}_{cs}$ *and* $\bar{s}_{cs} < s < s_{c2}^R$.

Counterintuitively, although uninformed consumers can enjoy a reduced return shipping cost freely under the RRSI policy, the consumer surplus declines. To understand this, note that the retailer actually not only transfers the insurance premium to consumers but also extracts the uninformed consumers' elevated surplus stemming from the insurance benefit by charging a premium "bundle price" higher than the sum of the premium and product benchmark price. Correspondingly, some informed consumers switch to not purchasing when RRSI is introduced. In this case, RRSI actually hurts the consumer surplus. In contrast, consumer surplus may be better off with CRSI irrespective of the extra expense. This is because the retailer lowers the product price such that more informed consumers purchase the product without return. Uninformed consumers purchasing products only are also better off. Thus, CRSI could achieve a "win-win-win" goal for the insurer, retailer, and consumers (see Figure 5).

Proposition 4 suggests that the insurer should pay more attention to CRSI, which can benefit not only itself but also

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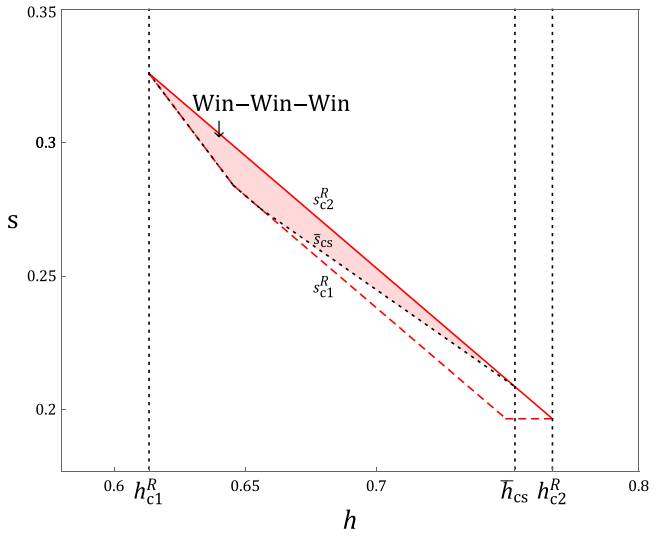


FIGURE 5 Customer–return shipping insurance (CRSI) ($K = 2$): win-win-win region. *Note*: $\theta = 0.05, \alpha = 0.5, c = 0.2, \gamma_N^H = 1, \gamma_N^L = 0.9$, $\gamma_I = 0.8$.
[Color figure can be viewed at wileyonlinelibrary.com]

the retailer and consumers although informed consumers will never be interested in it. Meanwhile, consumers should caution that *"there is no such thing as a free lunch."* The retailer offering RRSI for free may actually transfer the premium cost to consumers by charging a higher product price. On the contrary, CRSI can be a more economical tool for consumers to mitigate their concerns about product fit uncertainty.

With the coronavirus pandemic spreading in early 2020, retailers' online year-over-year revenue growth is 129% in American and Canadian e-commerce orders and 146% in all online retail orders as of April 21 (Columbus, 2020). Adopting RSI may provide an unprecedented opportunity for online retailers to improve profitability. Proposition 4 suggests that regulators such as the platform and government should consider promoting CRSI to benefit consumers, as well as the retailer and insurer.

6 | DISCUSSION

6.1 | Impact of a restocking fee

We have investigated RSI policies based on the retailer's full-refund return policy. In practice, some retailers adopt partial-refund policies, for example, by charging consumers a restocking fee. Also, some retailers use the full-refund policy but set some restrictions for consumers, including strict time limits for returns, return of the original packaging materials, and acceptance of only those products that show no visible signs of use.[18] We next extend our research by considering the retailer charging a restocking fee $r(r < h)$ for product returns. Proposition 5 presents the impact of the restocking fee on the RSI policy.

Proposition 5. *When the retailer charges a restocking fee $r(r < h)$, compared with the case of a full-refund policy, the choice of RSI policy does not change in nature. However, the RRSI policy is more appealing, while the opposite occurs for the CRSI policy.*

In the presence of a restocking fee r when returning an item, consumers have to pay r to the retailer in addition to the shipping fee s, and thus the consumers' return hassle cost increases from s to $s + r$. The restocking fee r offsets part of the retailer's return handling cost h, lowering it to $h - r$. Essentially, this revised game can be solved by following the main model with the consumers' return shipping cost set to $s + r$ and the retailer's return handling cost set to $h - r$. Correspondingly, the results hold qualitatively. In accordance with Proposition 2, the RRSI policy is more applicable while the parameter range for CRSI policy shrinks, due to the retailer's lower return handling cost and consumers' higher return shipping cost caused by the restocking fee.

Proposition 5 reveals that the results derived in the main model with the full-refund policy are robust to changes in the retailer's return policy. Our findings shed light on the impact of RSI policies on market players and provide practitioners with valuable guidance on adopting insurance policies, showing how the optimal choices depend on the consumers' return shipping cost and the retailers' return handling cost.

6.2 | Retailer offering RSI to consumers directly

If the retailer is allowed to offer RRSI directly (DRRSI), that is, not through the intermediate insurer, the retailer saves the purchasing cost of insurance premium λ_r from the insurer compared to the RRSI and pays only the compensation of c with the probability of $1 - \alpha$ to the uninformed consumers, that is, $(1 - \alpha)c < \lambda_r$, because informed consumers never return products and uninformed consumers do not return a product with the probability of α. In this case, RRSI offered by the insurer is not competitive anymore. The DRRSI equivalently separates informed consumers from uninformed ones, and the profit margins on these two segments are p_r and $\alpha p_r - (1 - \alpha)(h + c)$, respectively. Suppose CRSI is still offered through the insurer. Proposition 6 summarizes the optimal RSI policies and Figure A.11a in the Supporting Information depicts the regions for the optimal DRRSI and CRSI policies.

Proposition 6. *In equilibrium,*

(*i*) *if the retailer's return handling cost is relatively low $(h < h^{R'}_{r2})$, and the consumers' return shipping cost is medium $(h + c \equiv s^{R'}_{r1} < s < s^{R'}_{r2})$, the retailer offers DRRSI and charges a higher product price than in the benchmark case;*

(*ii*) *if H uninformed consumers are more sensitive to the uninsured regret than the insured regret $(\gamma_N^H > \gamma_I)$, the*

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return handling cost of the retailer is relatively high ($h^R_{c1} < h < h^R_{c2}$) (note $h^R_{c1} > h^{R'}_{r2}$), *and the consumers' return shipping cost is medium* ($s^R_{c1} < s < s^R_{c2}$), *then the insurer sets premiums such that the retailer offers CRSI and charges a lower product price than in the benchmark case;*

(*iii*) *otherwise, the insurer never offers DRRSI or CRSI.*

Proposition 6 indicates that the main results shown in Proposition 2 qualitatively hold. The only difference is that the region offering DRRSI is enlarged because the retailer's cost of providing this insurance goes down.

Furthermore, we explore the case where the retailer can offer CRSI directly to consumers (DCRSI). Unfortunately, the analysis is analytically intractable; therefore, we resort to a numerical study. Under DCRSI, the retailer determines both the product price and insurance premium, which is equivalent to the case where the retailer is the central decision maker for both the insurance premium and product price. In the numerical study, we vary h from 0 to 0.35 and s from 0 to 0.25 with a step of 0.001, which generates 88,101 problem instances. Figure A.11b in the Supporting Information shows that our findings in the main model still qualitatively hold. The only difference is that both regions for DCRSI and DRRSI expand due to the retailer's centralized decision-making.

6.3 | Private information on retailer's return handling cost

In the main model, we have assumed that all the information is public. In practice, the retailer may want to keep his return handling cost h as a business secret and not want to share it with the insurer. In this section, we analyze the equilibrium in such a case. The insurer will make the premium decisions based on its expectation of h. Suppose that the insurer anticipates the return handling cost to be high ($h^H > h$) and low ($h^L < h$) with probabilities δ and $1-\delta$, respectively. Then, the insurer makes its premium decisions based on its expected return handling cost $\delta h^H + (1-\delta)h^L \equiv H$ rather than h. If $H \neq h$, the insurer sets the insurance premium deviating from the optimal decision that maximizes its profit. The optimal RSI policies are summarized in Proposition 7 and the corresponding regions are depicted in Figure A.12 in the Supporting Information.

Proposition 7. *As long as the insurer's anticipation on h does not deviate too much, that is, $\bar{H}_1 < H < \bar{H}_2$, where $\bar{H}_1 < h < \bar{H}_2$, the optimal RSI policy does not change qualitatively. Specifically, in equilibrium,*

(*i*) *if the retailer's return handling cost is relatively low* ($h^{R''}_{r1} < h < h^{R''}_{r2}$) *and the consumers' return shipping cost is medium* ($s^{R''}_{r1} < s < s^{R''}_{r2}$), *the insurer sets premiums such that the retailer offers RRSI and charges a higher product price than in the benchmark case;*

(*ii*) *if H uninformed consumers are more sensitive to the uninsured regret than the insured regret ($\gamma^H_N > \gamma_I$), the return handling cost of the retailer is relatively high* ($h^{R''}_{c1} < h < h^{R''}_{c2}$) (note $h^{R''}_{c1} > h^{R''}_{r2}$), *and the consumers' return shipping cost is medium* ($s^{R''}_{c1} < s < s^{R''}_{c2}$), *then the insurer sets premiums such that the retailer offers CRSI and charges a lower product price than in the benchmark case;*

(*iii*) *otherwise, the insurer never offers RSI.*

Proposition 7 shows that our main findings qualitatively hold. Generally, when the insurer expects a higher return handling cost, that is, $H > h$, it would set a lower premium than in the main model. Accordingly, the retailer would set a lower selling price under RRSI but may set a higher product price under CRSI. Therefore, the negative AE effect is mitigated and the region for offering RRSI can be larger than in the main model. In contrast, the region for offering CRSI can be smaller because the negative DS effect is stronger due to the higher product price. The impact of H on the optimal RSI policy will be reversed when the insurer expects a lower return handling cost, that is, $H < h$.

6.4 | Beta distribution of product value

In this section, we examine whether our results hold under a different distribution. Beta distribution with parameters $a = 1$ and $b > 0$ is commonly used to model consumer preference on the product value v (Debo et al., 2005; Pan & Honhon, 2012; Sundararajan, 2004). Specifically, let $f(v) = \frac{x^{a-1}(1-x)^{b-1}}{\int_0^1 u^{a-1}(1-u)^{b-1}\,du} = \frac{(1-x)^{b-1}}{\int_0^1 (1-u)^{b-1}\,du}$ and $F(v) = 1-(1-v)^b$ with support [0,1] denote the probability density function and cumulative density function of customer valuations, respectively. If $b > 1$, more customers have low valuations because the distribution function is concave. In contrast, when $0 < b < 1$, the distribution function is convex, and so more customers have high valuations. The uniform distribution on [0,1] is a special case of the beta distribution when $b = 1$.

With the indifference point $\bar{v}$, the demand is $F(1) - F(\bar{v}) = (1-\bar{v})^b$ rather than $1-\bar{v}$. Due to the complexity, we use a numerical study for $b = 2$ and $b = \frac{1}{2}$ to test the robustness of our results. For $b = 2$, we vary h from 0 to 0.5 and s from 0 to 0.2 with a step of 0.0005, which generates 401,401 problem instances. For $b = \frac{1}{2}$, we vary h from 0 to 0.6 and s from 0 to 0.12 with a step of 0.0005, which generates 289,441 problem instances. Figure A.13 in the Supporting Information shows that, under beta distribution, our major results in the main model are robust: RRSI is optimal when the retailer's return handling cost is relatively low and consumers' return shipping cost is medium, and CRSI is optimal when consumers' uninsured regret level is high, the retailer's return handling cost is relatively high and consumers' return shipping cost is medium.

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7 | CONCLUSION

This paper develops a parsimonious model to investigate the optimal RSI policy for a monopolistic online retailer and an insurer considering the consumers' anticipated regret behaviors. We demonstrate that RSI can further improve a retailer's profitability, no matter whether a full-refund or partial-refund product return policy is employed. Interestingly, we first show that, when consumer regret behaviors are not considered, CRSI is never viable, but the retailer should offer RRSI if the return handling cost is relatively low and the consumers' return shipping cost is in an intermediate range. We then extend the model by incorporating the consumers' anticipated regret behaviors. By exploiting the consumers' high sensitivity to the uninsured regret, the retailer should offer CRSI if the return handling cost is relatively high and consumers' return shipping cost is intermediate. Counterintuitively, RRSI always leads to a lower consumer surplus, but CRSI can be a "win-win-win" proposition for the retailer, insurer, and consumers. Further, we uncover that the retailer has more incentive to adopt RRSI when consumers pay a restocking fee for product returns.

Theoretically, this work contributes to the literature on product return by revealing that retailers can raise their profit by offering RSI in addition to any product return policy. For example, retailers offering MBG can introduce a new return policy, "MBG + RSI," to enhance profitability. Complementing prior studies on RSI, our work shows the viability of both RRSI and CRSI policies.

Our findings potentially explain the prevalent RRSI and CRSI policies offered by retailers on platforms such as JD.com and Taobao.com. Our results have important implications for practitioners. Major retailers tend to adopt RRSI because they can repackage and resell returned products more economically. In contrast, it is more costly for small sellers to handle returned products; therefore, CRSI may be the best policy for them. Our results also suggest that retailers and insurers can advertise a cheap insurance premium and expensive product return shipping cost to raise the consumers' uninsured regret level so as to induce more consumers to purchase CRSI. We should caution that the presence of the RSI policies may be a joint effect of multiple factors. For example, small retailers may want to use a lower price to attract more consumers, and/or they may be unable to pay the insurer a large amount of premium fee to use RRSI ahead of time.

It is worth noting that some membership services such as Amazon Prime[19] and Best Buy Totaltech[20] membership include a free product return benefit within certain days since purchase, which allows consumers to return products with a free return shipping fee and waiver of restocking fee in addition to other benefits. RSI is different from these membership programs in the following two aspects. First, RSI is only for a return shipping fee for one transaction. RRSI is freely provided to all consumers by the retailer and all consumers can purchase CRSI with a low premium compared to the product price. In contrast, membership programs like Amazon Prime provide a bundle of benefits including fast/free delivery, prime video, amazon music, and prime gaming over a year. The membership fee is relatively high (e.g., $119 for Amazon Prime annual membership). Consumers make a long-term and complicated decision on whether to join the membership program. Second, the members' return-related costs will be covered by retailers while the compensation under RSI is covered by a third-party insurer. It will be interesting to explore whether a retailer should offer a membership program or RSI only in a future study.

Last, we point out some other interesting directions for future research. First, we investigate the optimal RSI policy using a one-shot game. In a multiperiod game, one can extend our work and explore the impact of other types of consumer regret, for example, the regret caused by repeated purchasing or switching between buying or not buying insurance. Second, we consider a monopolistic retailer's RSI policy adoption decision. It will be interesting to extend the model to a setting with competing retailers on an online platform. Third, we investigate the optimal RSI policy for a given product return policy. A model based on endogenous return policy would generate rich insights on how retailers' return policies and RSI policy interplay. Fourth, one can extend our research by allowing consumers to buy insurance from an independent vendor. Fifth, our work provides a theoretical foundation for future empirical work. It will be worthwhile to collect data from online platforms to quantify the benefit of RSI through an empirical study.

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ENDNOTES

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[6] http://tb.cn.hn/AZ8, accessed on January 10, 2020.
[7] http://tb.cn.hn/AZ9, accessed on January 10, 2020.
[8] Alternatively, we use beta distribution in the model extension, which shows that the major results are robust.

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[9] https://www.6pm.com/return-questions, accessed on February 21, 2020.

[10] The information on return shipping fee s is public for the retailer and insurer, as the shipping fee between any two destinations is publicly listed on the website of the delivery service company.

[11] The retailer may keep the information on h as a business secret and not share it with the insurer. In the main model, we assume the insurer can expect the exact h value, a model of private h information is considered in the extension, which shows that the major results are robust.

[12] According to the RRSI contract, the retailer cannot choose to provide RSI only for some consumers. Refer to Figures A.4 and A.6 in the Supporting Information.

[13] To facilitate analysis and exposition, we assume that the CRSI premium λ_c is identical to all consumers as in Zhang et al. (2022). Our results will qualitatively hold if the insurer can charge a different insurance premium based on consumers' product return history.

[14] We assume that consumers are not concerned about the product quality over time; therefore, we do not consider product warranty insurance, which is for long-term benefit and more expensive than RSI offering one-time benefit.

[15] http://tb.cn.hn/AZA, accessed on August 4, 2020.

[16] https://tb.cn.hn/BQvand https://tb.cn.hn/BQS, accessed on August 4, 2020.

[17] The CRSI policy is similar in Case 1 and Case 2. Hereinafter, we only use figures to illustrate CRSI policy under Case 2 for ease of exposition.

[18] See the example of AFA stores at https://www.afastores.com/return-policy.html, accessed on July 26, 2020.

[19] https://www.amazon.com/amazonprime, accessed on November 8, 2022.

[20] https://www.bestbuy.com/site/electronics/totaltech/pcmcat1629315977983.c?id=pcmcat1629315977983, accessed on November 8, 2022.

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